

Enterprise Knowledge Copilot — Data Sources & Data Engineering

Project: Enterprise Knowledge Copilot
Total Records Ingested: 645 chunks | **PII Leaks:** 0 (verified)

1. Data Sources Overview

#	File	Format	Description	Records
1	nexacorp_handbook.txt	TXT	HR policies, IT guidelines, SLA definitions, data classification rules, error code references	~40 chunks
2	nexacorp_vpn_auth_runbook.txt	TXT	IT troubleshooting procedures mapped to specific error codes (ERR-AUTH-, VPN-CERT-)	~25 chunks
3	nexacorp_tickets.csv	CSV	Historical IT incident reports with ticket IDs, priorities (P1-P4), error codes, resolutions	236 rows
4	nexacorp_org_chart.csv	CSV	Entity-relationship triples mapping employees to systems (e.g., Marcus Thompson → AUTH-GATEWAY)	~50 triples

These four sources cover the three pillars of enterprise knowledge: **Policy** (Handbook/Runbook), **History** (Tickets), and **Ownership** (Org Chart).

2. Data Engineering Pipeline (6 Steps)

Step 1: Raw File Extraction

Text files are read as raw strings. CSV files are parsed row-by-row using `csv.DictReader`, with each row flattened into a single searchable text string.

Step 2: PII Sanitization

A two-stage redaction pipeline strips sensitive data before indexing:

- **Stage 1 (Regex):** Detects and masks emails and phone numbers with `[REDACTED_EMAIL]` / `[REDACTED_PHONE]` tokens.
- **Stage 2 (NER):** spaCy's `en_core_web_sm` model detects PERSON entities. Names are sorted by length (longest first) to prevent partial leaks — redacting "Dr. Sarah Jenkins" before "Sarah" ensures no last names are exposed.

Step 3: Text Chunking

Unstructured text is split using a rolling window: **800 characters** per chunk with **150 character overlap**. The 800-char window keeps content within the embedding model's 512-token limit (~200 tokens), while overlap ensures boundary sentences appear in both adjacent chunks.

Step 4: Metadata Enrichment

Each document is tagged with structured metadata for filtered search:

Field	Source	Example	Enables
priority	Tickets	"P1"	"Show me P1 tickets"
status	Tickets	"Resolved"	Status-based filtering
error_code	Tickets	"ERR-AUTH-9092"	Error code lookups
system	Auto-detected	"AUTH-GATEWAY"	System-specific queries

Step 5: Knowledge Graph Construction

Org chart triples are loaded as directed edges in NetworkX (`G.add_edge(entity, target, relation=relationship)`). Additionally, automatic entity co-occurrence mining discovers implicit relationships by linking NER-extracted entities that appear in the same CSV row.

Step 6: Triple Indexing

Processed chunks are simultaneously committed to three independent indices:

Index	Technology	Storage	Purpose
Vector Index	ChromaDB + bge-small-en-v1.5	<code>.index/chroma/</code>	Semantic similarity search
Keyword Index	BM25Okapi	<code>.index/bm25.pkl</code>	Exact term matching
Graph Index	NetworkX DiGraph	<code>.index/graph.pkl</code>	Relational traversal

3. Data Quality Challenges Resolved

Challenge	Resolution
Malformed CSV rows with unquoted commas	Custom pre-processing script to detect and quote offending fields
Partial PII leaks ("Dr. [REDACTED] Jenkins")	Sort entities by string length (longest first) before redaction
Missing metadata fields in ticket rows	Graceful conditional injection with <code>.strip()</code> handling